

Of course, a code smell isn't necessarily a problem. Ultimately, whether to address or ignore a code smell is a judgment call for you to make.

Duplicate Code

The most common code smell is *duplicate code*. Duplicate code is any source code that you could have created by copying and pasting some other code into your program. For example, this short program contains duplicate code. Notice that it asks how the user is feeling three times:

```
print('Good morning!')
print('How are you feeling?')
feeling = input()
print('I am happy to hear that you are feeling ' + feeling + '.')
print('Good afternoon!')
print('How are you feeling?')
feeling = input()
print('I am happy to hear that you are feeling ' + feeling + '.')
print('Good evening!')
print('How are you feeling?')
feeling = input()
print('I am happy to hear that you are feeling ' + feeling + '.')
```

Duplicate code is a problem because it makes changing the code difficult; a change you make to one copy of the duplicate code must be made to every copy of it in the program. If you forget to make a change somewhere, or if you make different changes to different copies, your program will likely end up with bugs.

The solution to duplicate code is to *deduplicate* it; that is, make it appear once in your program by placing the code in a function or loop. In the following example, I've moved the duplicate code into a function and then repeatedly called that function:

```
def askFeeling():
    print('How are you feeling?')
    feeling = input()
    print('I am happy to hear that you are feeling ' + feeling + '.')

print('Good morning!')
```
